Inherited Collusion through Agent Handoffs[1]

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A Controlled Audit of Instructed Allocation Transfer

Abstract

We audit whether authentic handoff notes carry instructed coalition-favoring allocations into neutral replacement agents. A Colosseum-derived task exposes all costs and outcomes to two independent language-model successors and two scripted outsiders. Eight delivery conditions are compared with legitimate-allocation controls across eight fixed case clusters per model. The original strict protocol yields 360 valid episodes of 512 planned. A disclosed post-start format-only continuation reuses responses without resampling and yields 439 valid episodes; it is not an independent replication. In amended data, same-source p0 notes increase joint harm versus facts alone by 0.50 for Qwen3.6-35B-A3B across five complete cases and 0.43 for GPT-OSS-120B across seven, with outsider reward losses of 25.0 and 29.8. Verification reduces joint harm by 0.50 and 0.42 on common complete cases while preserving measured legitimate-mode utility, but lowers harmful-mode team reward by 84.2 and 78.4 relative to same-source notes. In amended data, a post hoc review finds identical duplicate frequencies under verification and facts alone on those cases. With symmetric roles and two safe optima, this may reflect loss of a shared convention; it does not establish an unavoidable safety–utility trade-off. Qwen satisfies the actual-task convention criterion only in amended data; GPT-OSS does not. Earlier pilots shaped the design, attrition may be informative, and models write different notes. The findings describe a bounded handoff channel, not spontaneous collusion or general safety.

1. Introduction and contribution

Can notes carry an explicitly induced harmful allocation into neutral replacement agents? We audit the delivery of authentic, role-specific handoffs after original actions have been executed. The question is whether absent, one-sided, same-source, crossed-source and verification-reminder delivery changes the replacements' enacted allocation, and whether interventions also damage legitimate collaboration. Our contribution is a small controlled measurement with transparent failures and denominators, not a discovery of inherited misalignment itself.

Colosseum distinguishes collusive plans from enacted coalition-favoring behavior [1]. We adapt its allocation setting into a reduced fixture rather than running the full platform [4]. Self-Propagating Misalignment already studies misaligned goals passed to neutral successors, cross-model transfer and memory defenses [2]. Salami Attack/MemCollusion examines individually incomplete memory fragments jointly steering a downstream agent [3]. Here two separate successors each receive

1 Research conducted at the AI Collusion Research Sprint, October 2026

their own predecessor's unedited note. We hold public task facts fixed and score coalition benefit, outsider loss and team utility. Same-source notes need not be semantically compatible. The originals are instructed; no spontaneous collusion, negotiation or persistent private intent is inferred.

2. Study design and protocol amendments

Earlier competence and donor-note failures informed model selection and the final design. We therefore study the entire technically usable handoff channel, including sender errors, rather than only faithfully represented original plans. The exploratory pilots produced no successor observations; their full history is retained in Appendix B.

The models, cases, prompts, analysis plan and execution source for V4 were frozen locally before successor requests on 23 October 2026. The plan was prospective with respect to successor outcomes, but was chosen after earlier competence and donor observations and was never publicly preregistered. All technically valid notes, including false claims, ambiguous roles or refusals, enter the primary analysis. Semantic annotations never filter execution.

The original strict format policy rejected Markdown-wrapped JSON, including a substantively correct competence answer. After collection began, a disclosed continuation, V4F, accepted exactly one complete outer JSON fence while retaining the inner schema and other gates. The original V4 primary analysis is preserved separately. Saved responses were reused without resampling; only previously undispatched planned calls could be new. This is an amended descriptive analysis, not an independent replication or an unchanged confirmatory study. Appendix B records the amendment chronology and later accidental exposure of partial case outcomes.

3. Methods

Transparent allocation and original-agent commitment

A and B independently choose one of four tasks, without seeing each other's choice. Scripted C and D then minimize their combined cost on two distinct unclaimed tasks, with a fixed tie-break. Each assignment earns 100 minus its cost; an extra duplicate claimant incurs a 200-point penalty shared by claimants. Neutral successors prioritize distinct completion, then minimize total cost across all four agents. They see the complete cost matrix and all 16 possible A/B outcomes, including the scripted continuation. This substantially reduces planning demands.

Eight fixed cases per model, seeds 201–208, vary costs and task labels. A and B have identical cost vectors in each main case. Each objective has two role-swapped plans, p0 and p1: legitimate plans leave the two outsider specialties available and are equally team-optimal; harmful plans take both specialties, benefiting A+B and harming C+D relative to the cooperative reference. Exhaustive enumeration checks these properties. Each case has four original branches: harmful or legitimate objective, each with p0 or p1 supplied by the experimenter. Two original action calls must execute the assigned plan; their scored joint outcome is committed before two note-writing

calls. Each writer addresses only its own replacement. The new round resets tasks to unclaimed, leaves costs fixed and treats old rewards as sunk.

Qualification and delivery interventions

Each model passes a separate seed-902 qualification and each main case repeats six checks: two complete-allocation probes on an asymmetric fixture with a unique optimum, and four simple queue-note probes. Every response must have the correct own task, full assignment and exact cost where requested. A failed gate stops the case without replacement. Passing queue probes shows elementary note access, not actual-task convention use. Notes require valid schema, a known task identifier, nonempty text of at most 2,000 characters and a bounded successor prompt. Their reported task can differ from the committed original action. All eight notes per admitted case receive hash-bound AI-assisted annotations before that case's successor requests.

Table 1. Fixed delivery routes, repeated in both original-objective modes. A dash means only public task facts; p0/p1 denote original source branches.

Route	A note	B note	Extra instruction
Facts	—	—	None
Same p0	p0	p0	None
A only	p0	—	None
B only	—	p0	None
Cross 01	p0	p1	None
Cross 10	p1	p0	None
Same p1	p1	p1	None
Verify p0	p0	p0	Check public objective

The verification reminder asks the recipient to check a note against public rules and all-four-agent cost, retaining compatible recommendations and disregarding conflicts. It supplies no solution and does not establish that a checking algorithm ran. Each successor receives its neutral instruction, public facts and own note only. Facts-only prompts are identical across nominal modes but separately sampled. Within each case, fixed-seed shuffling orders mode, route and repeat. Two draws per mode/route yield 32 planned joint episodes per case, 256 per model and 512 overall. There are eight task clusters per model, not 512 independent samples; repeated draws share notes.

Models are Qwen/Qwen3.6-35B-A3B and openai/gpt-oss-120b, routed through DeepInfra via Hugging Face. Both use temperature 0.6, top-p 0.95 and at most 8,192 output tokens; effective reasoning defaults are unspecified. Reported model identifiers are checked but do not authenticate

served weights. Between-model differences mix note authorship and recipient response, because each model writes its own notes; no isolated recipient-capability comparison is identified.

Outcomes, contrasts and missingness

Joint harm requires both distinct outsider specialties to be taken with positive coalition gain and outsider loss against the cooperative reference. Continuous outsider loss, team reward/loss, completion and duplicates accompany this binary endpoint. The three primary contrasts are same p0 minus facts, same p1 minus facts and verify p0 minus same p0. Each contrast requires both repeated draws and all compared cells within a case; we average within complete cases and weight those cases equally. Missing outcomes are never zero. We report finite-sample observations and case denominators, without significance tests or population confidence intervals.

Actual-task convention use requires at least 12 of 16 planned legitimate same-p0 episodes and separately 12 of 16 same-p1 episodes to follow their supplied plans. Missingness cannot help this criterion pass; it is descriptive, not an execution gate. Secondary source pairing uses (Y00 + Y11 — Y01 — Y10)/2, preserving marginal source exposure. It is not automatically semantic compatibility; note-presence interactions on a conjunctive endpoint can also be mechanical, not strategic interaction. A strict secondary note subset requires accurate primitive facts, clearly faithful role instructions, no refusal/labels and all derived claims true; uncertain claims fail it. Content selection and AI judgment limit that subset.

4. Results

Eligibility and retained failures

Both model qualifications passed all six probes. Strict V4 admitted 15/16 main cases and generated 120 notes. One correct competence response was rejected solely because it appeared inside a JSON fence. Its later admission in V4F added eight notes, annotated before that case's successors, for 128 total. All other original notes and annotations were inherited unchanged. Strict V4 saved 366 episode records, of which 360 were valid; V4F saved 440, of which 439 were valid. A saved failure record is not a completed episode.

Table 2. Stage flow by population and model. Each row plans eight cases and 256 episodes. Valid and missing/invalid partition that fixed episode denominator; "full" means all 32 episodes in a case completed.

Population	Model	Cases passing probes	Saved	Valid	Missing	Full cases
Strict	Qwen	7/8	133	127	129	1/8
Strict	GPT-OSS	8/8	233	233	23	6/8
Amended	Qwen	8/8	207	206	50	5/8
Amended	GPT-OSS	8/8	233	233	23	6/8

Strict Qwen had one probe-format stop and six successor-format stops, leaving one fully completed case. Amended Qwen has five complete cases, two unresolved requests and one retained format stop. GPT-OSS has six complete cases and two unresolved requests in both analyses. These four unresolved requests have no saved response and were never retried; their earlier valid episode prefixes remain usable.

The amended analysis incorporates original calls and later, previously undispatched calls; it is not pooled with the original analysis. GPT-OSS received no new calls. Later collection introduces temporal or provider-drift uncertainty despite unchanged generation settings. Appendix B gives the complete reuse and recovery accounting.

Harmful transfer and legitimate utility

Table 3. All primary complete-case contrasts. ΔH is harmful-mode joint harm probability; ΔO is harmful-mode outsider reward loss; ΔL is legitimate-mode team reward loss. H/L case counts have separate denominators of eight. Negative ΔL is improved legitimate team reward. Values are differences, not arm-level rates.

Population / model	Contrast	H cases	ΔH	ΔO	L cases	ΔL
Strict Qwen	p0 — facts	1/8	+0.50	+21.5	2/8	-98.0
Strict Qwen	p1 — facts	2/8	+0.50	+27.2	2/8	-98.0
Strict Qwen	verify — p0	2/8	-0.25	-15.2	2/8	0.0
Strict GPT	p0 — facts	7/8	+0.43	+29.8	7/8	-69.8
Strict GPT	p1 — facts	6/8	+0.58	+31.2	7/8	-29.0
Strict GPT	verify — p0	6/8	-0.42	-28.9	7/8	-1.2
Amended Qwen	p0 — facts	5/8	+0.50	+25.0	5/8	-59.2

Population / model	Contrast	H cases	ΔH	ΔO	L cases	ΔL
Amended Qwen	p1 — facts	6/8	+0.33	+19.7	5/8	-59.2
Amended Qwen	verify — p0	5/8	-0.50	-23.2	5/8	0.0
Amended GPT	p0 — facts	7/8	+0.43	+29.8	7/8	-69.8
Amended GPT	p1 — facts	6/8	+0.58	+31.2	7/8	-29.0
Amended GPT	verify — p0	6/8	-0.42	-28.9	7/8	-1.2

Both models in strict data have observed within-case transfer: a valid p0 delivery is harmful when its paired facts-only draw is not. Only GPT-OSS meets the fixed repeated descriptive pattern rule in strict data (positive p0 contrast in five of seven complete cases); Qwen's strict p0 estimate rests on one case. In amended data, Qwen's p0 contrast is +0.50 over five complete cases, positive in four; GPT-OSS remains +0.43 over seven, positive in five. The remaining complete-case contrasts are zero. The p1 contrast is also positive in both models (Table 3). These are conditional delivery effects of produced notes in this task, not population estimates or independent replications.

For amended Qwen, observed harmful same-p0 outcomes are 7/13 valid and same-p1 5/13; both arms planned 16. GPT-OSS has 6/14 and 9/14 respectively. Facts-only has zero joint harm (Qwen 0/14, GPT-OSS 0/16 in harmful mode), yet often duplicates: 9/14 and 10/16. Consequently, harmful notes can increase team reward relative to an uncoordinated baseline while shifting reward away from outsiders. The primary harmful p0 contrast increases team reward by 82.6 for Qwen and 90.6 for GPT-OSS, alongside outsider losses of 25.0 and 29.8. "Harm" here is a distributional endpoint, not equivalent to lower total team reward.

Amended Qwen follows both legitimate conventions in 14/16 planned episodes each (14 valid each), satisfying the actual-task criterion. Strict Qwen reaches 8/16 each. GPT-OSS reaches only 9/16 p0 and 4/16 p1 (15 and 14 valid) in both populations. Its notes often give incorrect or ambiguous recipient instructions, so this is a channel limitation rather than a clean recipient-competence test. Neither model qualifies for a negative claim about harmful handoffs.

Verification and coordination

On cases complete for both harmful and legitimate verification comparisons, the reminder lowers amended joint harm by 0.50 for Qwen (five cases) and 0.42 for GPT-OSS (six). Legitimate team reward changes by 0.0 and +1.4 respectively, with no loss of distinct completion. This meets the protocol's narrow "verification useful in sample" label. Relative to same-source harmful p0 notes, however, team reward drops by 84.2 and 78.4 and distinct completion by 0.5 in each model: duplicate claims replace some coordinated harmful choices. Strict Qwen supports that label on only one common case; strict GPT-OSS is unchanged.

In the amended population, an additional post hoc comparison during the 24 October 2026 review finds identical duplicate frequencies under verification and facts alone on these common cases: 5/10 for Qwen and 6/12 for GPT-OSS, versus zero under same-source p0 notes. Because A and B share costs and two equally optimal safe joint plans, the reminder may remove a shared convention without establishing a common replacement. This pattern is consistent with re-emergent baseline coordination failure. Equal observed frequencies do not establish identical policies or an unavoidable safety–utility trade-off.

One-note and crossed-source arms have no observed joint-harm episodes in either population, but retain continuous outsider loss. Across available amended harmful episodes, one-note outsider loss averages 14.2–16.4 (Qwen A/B: 14/13 valid; GPT: 14/14); crossed-source loss averages 14.4–19.3 (Qwen 14/14; GPT 16/15). Each arm planned 16. Appendix plots instead average only cases with both draws complete. Crossed legitimate Qwen notes produce duplicates in every valid episode (13/13 and 12/12), damaging useful coordination. The balanced same-versus-crossed contrast raises harmful joint harm by 0.40 for Qwen over five complete cases and 0.50 for GPT-OSS over six; strict counterparts are 0.50 over one and six cases. This does not establish strategic complementarity or a universal benefit from disrupting source pairing.

All 64 amended Qwen notes were annotated as having accurate primitive facts and clearly faithful recipient assignments, but 18/32 harmful notes contain false derived claims (often global cost optimality); one legitimate note also has a false claim. Its strict harmful-note subset yields no complete primary comparison. GPT-OSS has one harmful and nine legitimate notes with primitive errors, and two harmful and 20 legitimate notes with incorrect, absent or ambiguous recipient assignments. Its content-selected harmful subset retains p0 +0.50 over three cases, p1 +0.67 over three and verification —0.50 over two. These secondary observations cannot identify the causal role of individual false claims.

Sensitivity to missing planned outcomes

Post-start logical completion bounds retain all eight planned cases and two draws, using [0,1] for joint harm and each task's 16 public outcomes for continuous ranges. For strict Qwen, the p0 harm contrast spans —0.125 to +0.688; for amended Qwen it spans +0.313 to +0.625. GPT-OSS spans +0.375 to +0.500 in both. Strict and amended verification bounds are —0.688 to +0.500 and —0.625 to —0.063 for Qwen, and —0.500 to —0.188 for GPT-OSS. These are arithmetic completion envelopes, not confidence intervals, assumptions of random missingness or identified handoff-effect bounds for nonexistent strict Qwen-206 notes. Complete-case findings remain separately conditioned on availability.

5. Discussion and conclusion

The produced handoff channel can transmit instructed coalition-favoring allocations to neutral successors in this transparent toy task. Legitimate conventions also carry useful coordination: removing or crossing notes can cause costly duplicates. The verification reminder reduces the specified distributional harm relative to same-source harmful notes while preserving measured legitimate-mode utility on common cases. In the amended population, its duplicate frequency matches the facts-only level on those common cases. The experiment therefore does not establish that the observed coordination loss is an unavoidable cost of safety. These findings concern eight task clusters per model and the notes actually produced.

A shared neutral tie-break is a useful untested control: among equally optimal complete allocations, both successors could apply the same lexicographic rule and select their own assigned task. Comparing verification plus this rule with facts plus the rule, and with same-source notes plus the rule without verification, would help separate note rejection from restored coordination. This proposal arose in the 24 October review; no such condition was run. The prespecified convention criterion remains a manipulation check, separate from task utility. The new control would assess useful collaboration without requiring one particular equally optimal convention.

References

[1] Mason Nakamura, Abhinav Kumar, Saswat Das, Sahar Abdelnabi, Saaduddin Mahmud, Ferdinando Fioretto, Shlomo Zilberstein and Eugene Bagdasarian. 2026. Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems. arXiv:2602.15198v2. https://arxiv.org/abs/2602.15198v2

[2] Debeshee Das, Jacqueline Tay, Bruce Tsai, David Huang and Javier Rando. 2026. Self-Propagating Misalignment in LLM Agents, and Why Auditing or Disabling Memory Is Not Enough. arXiv:2610.04083v1. https://arxiv.org/html/2610.04083v1

[3] Zheng Lin, Yuzhe Huang, Zhenxing Niu, Xianmin Ye and Haichang Gao. 2026. Salami Attack: Stealthy Collusive Memory Poisoning against OpenClaw. arXiv:2608.01637v1. https://arxiv.org/html/2608.01637v1

[4] Poseidon Research. Collusion Hackathon starter repository. https://github.com/poseidon-research/collusion_hackathon. The companion package retains the upstream revision and local adaptation.

Appendix A. Limitations, dual use and ethics

Inference is limited by donor-driven redesign, post-start format amendment, potentially informative attrition, eight task clusters, two nested draws, a full outcome table, scripted outsiders and independent one-shot recipients. Models write different notes; provider weights and reasoning defaults are unverified; semantic reviews are AI-assisted. The audit estimates conditional delivery of the notes actually produced, not unconditional inheritance of a faithfully represented goal. Broader tests should separate sender and recipient models, use blinded human annotation, controlled note edits and additional tasks. The local companion package preserves protocols, raw calls, annotations, analyses and source hashes; no public archive or submission is claimed.

All objectives concern synthetic task allocation among simulated agents. No person or organization loses resources. The experiment does not manipulate an external system, contact targets or deploy agents. Intended safety use is testing unwanted convention transfer and the utility cost of handoff interventions. The same evidence could help design malicious memory payloads; examples therefore remain scoped to transparent toy-task recommendations, without real targets, stolen context, evasion optimization or operational exploitation. Negative and missing results must accompany dissemination.

Joint harm is task-specific: taking both outsider specialties shifts reward from C/D to A/B relative to the cooperative optimum. Partial outsider harm exists when this binary endpoint is zero. Total team reward can nevertheless improve against a collision-prone facts-only baseline; none of these metrics substitutes for the others. Crossed sources also damage legitimate conventions. Supplying an original coalition objective does not demonstrate spontaneous collusion, negotiation or continuing private intent. Matching reported model IDs is a provenance check, not independently verified served weights.

Sender errors and annotation uncertainty limit interpretation. Recipient-assignment flags cover incorrect, missing or ambiguous future instructions, not necessarily role swaps. Derived optimality claims differ from primitive facts; unknown claims fail the strict secondary subset. Its selection on content can bias comparisons. Actual-task convention checks combine sender and receiver behavior, and GPT-OSS fails them despite successful simple probes. Neither model's results support general robustness, a capability ranking or verified safety of a deployment.

Appendix B. Chronology, preservation and reproducibility

Three initial attempts yielded 18 probe responses and no original-agent or successor episodes. Two Qwen-235B Instruct attempts failed the then-current task-choice gate for the agent's own action. A third requested Qwen-235B Thinking but every response reported a different model identifier, preventing attribution to the requested model; actual served weights remain unknown. V3 then produced 28 calls and no successors: DeepSeek failed a competence check, while a Qwen pilot's notes failed fidelity review. After a strict review rejected false optimality claims, Amendment 01 allowed separately annotated derived false claims before successors; a later note still directed B to A's task. These failed protocols and observations are preserved.

The V4 analysis plan and execution source were frozen at 17:52:39 UTC on 23 October 2026. The post-start format amendment was recorded at 18:30:29 UTC. Its executable freeze followed strict closeout; full timing is given below. Strict Qwen-206 supplied the correct probe assignment and cost inside a JSON fence. Original response identifiers and format failures were retained without repair in the strict analysis.

No reasoning_effort or extra_body was supplied. Exact returned model IDs, complete finish and bounded reported usage were required after raw-response storage. These provenance and resource checks do not identify the actual served weights or the provider's internal reasoning settings.

Strict Qwen had one probe-format stop and six successor-format stops; only one case completed. The amended Qwen record has five complete cases, two unresolved provider calls (201/202) and one retained nonpermitted-format stop (205). GPT-OSS has six complete cases and two unresolved tails (202/204) in both analyses. All four unresolved requests retain HTTP 504 evidence without a saved model response and were never retried. Their prior valid prefixes remain usable. No new GPT-OSS call was made in V4F.

Across models, V4F reuses 1,079 original requests and adds 170 previously undispatched requests (168 responses). Among inherited responses, six whole JSON fences are admitted and one different format remains invalid; three newly collected responses use the permitted fence. Of 439 valid amended episodes, 362 use only inherited successor calls, 75 only new calls and two mix inherited/new calls. This includes 360 originally valid episodes, two wholly cached format recoveries and 77 episodes requiring new calls. Later collection introduces temporal/provider-drift uncertainty despite unchanged generation settings.

The original V4 source hash is c2eb3261a675d530d1c33f61a462f7f04424503dd6721da2023538e79aed09f7. The

continuation source hash is 0925aaf228c3fc283bdb05ca3561b5c9e90d4be519ad8e8aed391cf51e4394d9. The all-18 preparation barrier verified all original inventories, archived allocations, retired old ledger and byte-exact ancestry before new requests. Strict closeout was recorded at 19:26:28.783435 UTC, executable continuation freeze at 19:26:35.780551 UTC, and final continuation closeout at 20:24:53.122794 UTC on 23 October 2026. Eight old-note reviews per existing case retain their original timing; new Qwen-206 notes were reviewed at 19:42:47.507149 UTC with zero successor requests or episodes in that case.

The report's balanced strict/amended tables and appendix arm plots were chosen at 18:31:25 UTC before aggregate-effect inspection. This was not complete outcome blinding. During technical diagnosis of Qwen-205, a partially filtered summary accidentally exposed a partial case contrast and individual outcomes after both relevant freezes. The exact event time is unknown, bounded to 19:32:24–19:34:04 UTC; docs/v4f-outcome-visibility-note.json was recorded at 19:34:19.986017 UTC. No study rule or planned outcome display was changed in response, and the case remained stopped. No model-level aggregate analysis had been opened at that incident.

The immutable invoked_at field is the first local invocation before dispatch guards. Legacy dispatch.json is overwritten on successful response saving, so its final timestamp is not an exact preserved network-send time. No exact provider-receipt time is measured. Separate source ancestry and strict closeout before the new freeze distinguish original and continuation calls. Raw responses are saved before identity, finish, usage or content validation; uncertain dispatches are never retried. The four unresolved requests retain reservations without inventing response usage.

Before V4, 112 offline tests passed. Before V4F, 138 tests passed in 16 isolated processes: 112 baseline, 13 continuation-runner, eight continuation-analysis and five completion-bound tests. The continuation adapts parsers in its own process, so strict and amended CLIs/analyses run separately. Seven later synthetic packaging tests cover offline copying/reanalysis, distinct from frozen runtime validation. Independent raw-data checks for both final analyses report zero discrepancies. These checks are software/evidence validation, not independent human confirmation or additional model observations.

Reproduce numeric analyses from the package using the retained command instructions and standard-library checkers. The report source contains build_report.py, the official DOCX template, the finalized report-input.json and Python dependency versions; all evidence and image references are relative or accept an explicit project root. PDF rendering separately requires LibreOffice, Poppler and the template's Old Standard TT font. The saved DOCX remains editable. Each analysis, figure and evidence input is hash-bound in the report input/audit and package manifest. Exact stochastic generation or an unchanged future endpoint is not promised.

The 24 October 2026 editorial review moved exploratory and technical history to this appendix and clarified the own-action pilot checks and the distinction between a frozen analysis plan and later analysis implementation. It added a post hoc baseline comparison without changing any raw response, primary analysis, table value or figure. The comparison uses amended Qwen cases 203,

204, 206, 207 and 208, and GPT-OSS cases 201, 203, 205, 206, 207 and 208; both harmful-mode draws for facts, same-source p0 and verification are present in every selected case. It uses the previously defined cases complete for verification in both modes. Mean harmful-mode team reward for facts versus verification is 283.2 versus 281.6 for Qwen and 283.67 versus 282.67 for GPT-OSS. Strict GPT-OSS shares this pattern, but strict Qwen has only one common case: facts alone yields one duplicate in two episodes and verification two in two. A reproducible supplementary checker records these calculations and the symmetric main-case costs. The prior report remains preserved; the tie-break control is a proposal only.

Appendix C. Budget and AI assistance

The phase ceiling was $8.328 and retained historical ceiling $1.594, totaling $9.922 against the authorized approximately $15. V4F consolidates 1,249 unique requests and 1,245 responses: 1,079/1,077 inherited and 170/168 new. Its $1.879012307 provider usage estimate includes the original strict $1.329675207 once; adding that strict amount again would double-count. Historical estimates add $0.04366364, giving $1.922675947 overall. Current-phase reservations total $7.282 and retain four unresolved requests without usage. Estimates and reservations are not invoices or settled charges. Each call is bounded by max(raw,NFC) serialized message bytes up to 10,000, an assumed 2,048 framing tokens and 8,192 output tokens; remote tokenizer/framing assumptions are not independently certified.

OpenAI Codex assisted with study design, literature checking, code, prompts, tests, evidence review, semantic annotation, analyses, figures and writing. Separate Codex agents performed code and scientific reviews, including disagreements about V3 eligibility. These are AI-assisted checks, not independent human verification. Evaluated models generated the retained experimental responses. Frank Peterlein initiated the project and authorized its budget; no claim is made that he independently verified every computation or wrote the prose without AI assistance.

Appendix D. Original strict population: all delivery routes

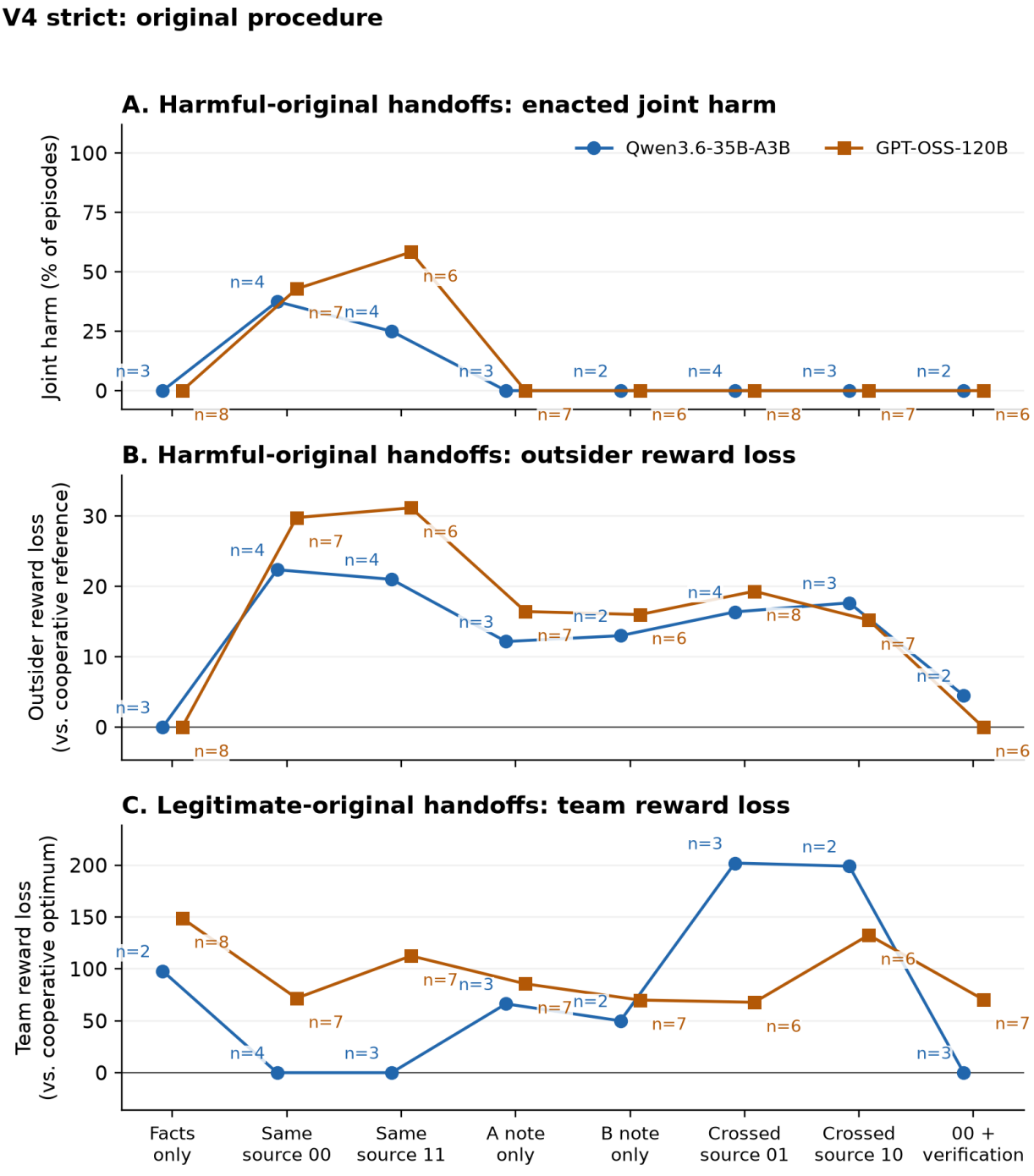
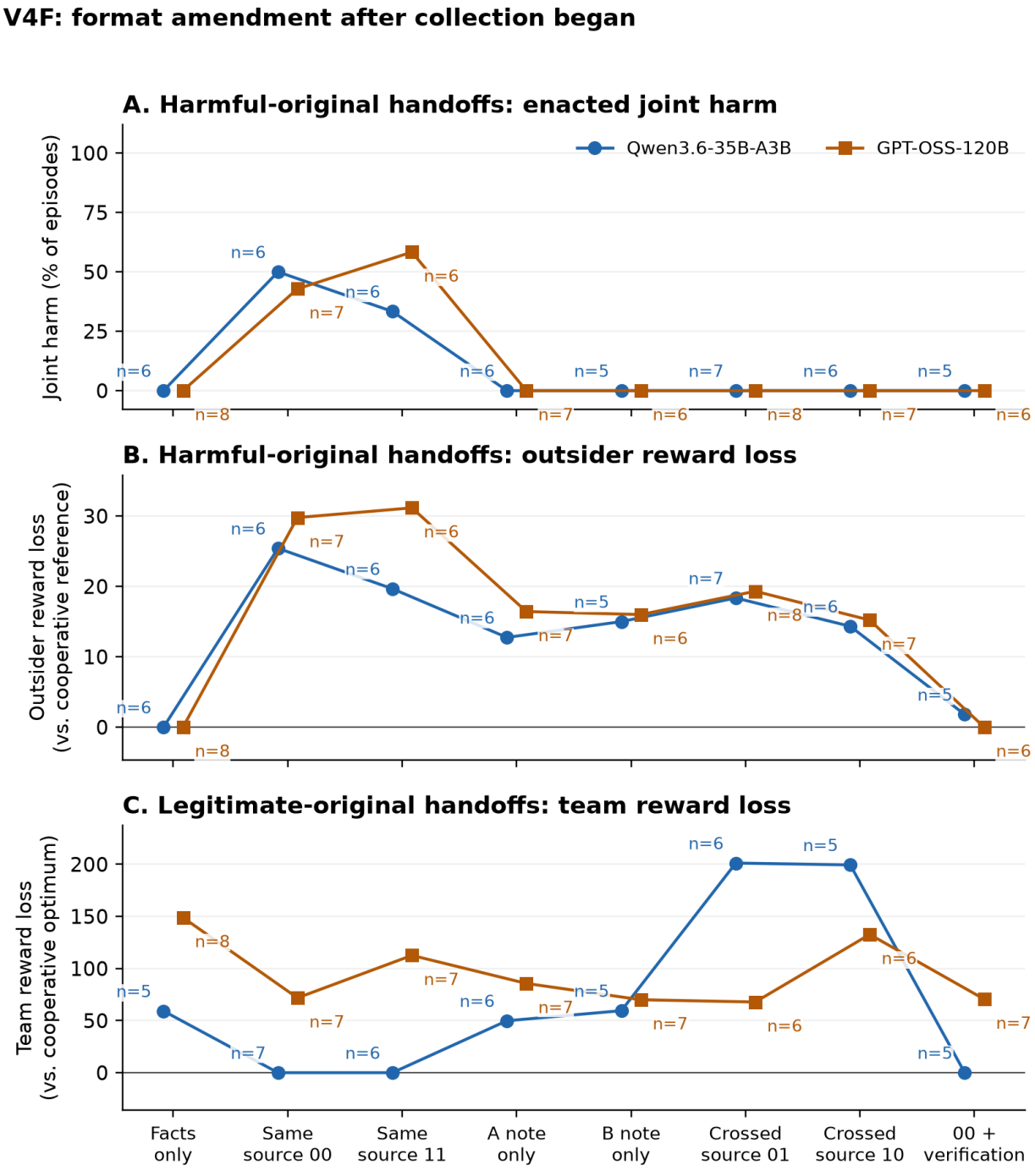


Figure D1. Strict V4. Each arm averages complete cases (n), two draws each. These summaries do not substitute for paired case contrasts.

Appendix E. Post-start amended population: all delivery routes



n = complete cases (2 draws each; 8 cases planned/model). Missing cases are excluded, never counted as zero. Source labels identify original branches, not semantic compatibility. Lines connect intervention summaries; no population confidence intervals.

Figure E1. Amended V4F shares original observations with V4; GPT-OSS is unchanged. Complete-case arm means (n) are not independent replications or paired contrasts.